Type MS · Does a geometric series have a sum to infinity?

Answer 1 of 4 · Question 7 part b · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

7(a)	$\frac{(x+1)}{(x-3)}=\frac{(4x-2)}{(x+1)}$ or $(x+1)^2=(x-3)(4x-2)$	M1
	$(x+1)^2=(4x-2)(x-3)\Rightarrow 3x^2-16x+5(=0)$	A1
	$(x-5)(3x-1)=0\Rightarrow x=5,\ \frac{1}{3}$	M1A1A1 (5)
(b)	$x=\frac{1}{3}\Rightarrow r=\frac{x+1}{x-3}=\frac{\frac{4}{3}}{\frac{-8}{3}}$	M1
	$r=-\frac{1}{2}$ ∴ convergent as $-1<r<1$ *	A1cso (2)
(c)	$S=\frac{a}{1-r}=\frac{-\frac{8}{3}}{1+\frac{1}{2}}=-\frac{16}{9}$ oe	M1A1 (2)
(d)	$\frac{S}{S_n}=\frac{a}{1-r}\times\frac{1-r}{a(1-r^n)}=\frac{1}{(1-r^n)}=\frac{256}{255}$	M1
	$255=256(1-r^n)$	
	$256r^n=1$ $\left(-\frac{1}{2}\right)^n=\frac{1}{256}$ oe $n=8$	dM1A1 (3) [12]

(a)M1	Form an equation using the given information about the terms
A1	Simplify their equation to a correct 3TQ, terms in any order Condone missing = 0
M1	Attempt to solve their 3TQ by any valid method. Must reach x = ... (at least one root) Calculator solutions: *Both* roots correct from a correct equation scores M1A1A1 Incorrect equation or incorrect roots scores M0A0A0
A1	One correct value of x
A1	Both correct values of x
(b)	
M1	Use either of their values of x, provided it is < 1, to find the corresponding value of r. No need to simplify
A1cso	Correct value of r **and** the conclusion including the reason
(c)	
M1	Use their value of r (not x) provided $-1<r<1$ (as found in (b) or here) and the formula for the sum to infinity to obtain a value for S
A1	Correct value
(d)	
M1	Obtain an equation in r and n. May use the formulae to cancel a or may sub values of a and r in the formulae for the LHS Must equate to 256/255 Value of r not needed for this mark so allow any value used.
dM1	Solve their equation of the form $r^n=...$ where $-1<r<1$ (r not x). May use trial and improvement or logs. This mark can be given if the equation and value of r are incorrect. Evidence of method needed if final answer is incorrect. If logs used condone log $(-1/2)^n$
A1	Correct value from correct working

Type MS · Does a geometric series have a sum to infinity?

Answer 1 of 4 · Question 7 part b · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	5	

Type MS · Does a geometric series have a sum to infinity?

Answer 2 of 4 · Question 4 part b · 1 mark

2025 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number		
4 (a) (i)	$\frac{ar^6}{ar^2}=\frac{40}{10}$	M1
	$r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$	M1
	$r=-\sqrt{2}$	A1
(ii)	$ar^2=10\Rightarrow a=\frac{10}{'2'}$	M1
	$a=5$	A1 (5)
(b)	$\lvert r\rvert>1$ so the series is not convergent.	B1FT (1)
		Total 6 marks

Part	Mark	Additional Guidance
(a)(i)	M1	For $\frac{ar^6}{ar^2}=\frac{40}{10}$ or $\frac{ar^2}{ar^6}=\frac{40}{10}$
	M1	For rearranging to find a value for r $r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ or $\left(\pm\sqrt[4]{4}\right)$
	A1	For the correct value of $r=-\sqrt{2}$ Accept $-\sqrt[4]{4}$
(a)(ii)	M1	For attempting to find a value for a using their r $ar^2=10\Rightarrow a=\frac{10}{'2'}$
	A1	For the correct value of $a=5$ Allow this mark from the use of $r=\sqrt{2}$
(b)	B1FT	$\lvert r\rvert>1$ so the series is not convergent. Follow through their value of r provided it is greater than 1.

Type MS · Does a geometric series have a sum to infinity?

Answer 2 of 4 · Question 4 part b · 1 mark

2025 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Does a geometric series have a sum to infinity?

Answer 3 of 4 · Question 8 part b · 1 mark

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

8(a)	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe $r = \left\{\frac{a(1+r)}{ar(1+r)} =\right\} \frac{288}{360} = \frac{4}{5}$ $a = \frac{360}{1 + "\frac{4}{5}"} = 200$ $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	M1M1 (B1B1on ePen dM1A1 M1A1 A1cso(B1 on ePen) [7]
8(a) ALT	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe $a\left(\frac{360}{a} - 1\right) + a\left(\frac{360}{a} - 1\right)^2 = 288$ $\Rightarrow a = 200$ $r = \frac{360}{"200"} - 1 = \frac{4}{5}$ $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	M1M1 (B1B1 on ePen) dM1A1 M1A1 A1cso(B1 on ePen) [7]
(b)	{S is convergent because} $\left\|\frac{4}{5}\right\| < 1$	B1 [1]
(c)	$S_\infty = \frac{'200'}{1-'\left(\frac{4}{5}\right)'} = 1000$	M1A1cao [2]
(d)	$\frac{200\left(1-\left(\frac{4}{5}\right)^n\right)}{1-\left(\frac{4}{5}\right)} > 978$ $\left(\frac{4}{5}\right)^n < \frac{11}{500}$ $n\,log\left(\frac{4}{5}\right) < log\left(\frac{11}{500}\right)$ $\Rightarrow n > 17.104\ldots$ $n = 18$	M1 dM1 ddM1 A1 [4]
		Total 14 marks

Type MS · Does a geometric series have a sum to infinity?

Answer 3 of 4 · Question 8 part b · 1 mark

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For either correct equation. Allow use of U_1 as the first term instead of a
	M1	For both correct equations. Allow use of U_1 as the first term instead of a
	dM1	For eliminating a from their equations, reaches r =... (Depends on previous mark)
	A1	For the correct value of r
	M1	Uses their value of r to find a value for a
	A1	For a = 200
	A1 cso	For the correct required expression of $\boldsymbol{U_n} = 200\left(\frac{4}{5}\right)^{n-1}$ (must have $\boldsymbol{U_n}$)
(a)ALT	M1	For either correct equation.
	M1	For both correct equations
	dM1	For eliminating r from their equations, reaches a =... (Depends on previous mark)
	A1	For the correct value of a
	M1	Uses their value of a to find a value for r
	A1	For the correct value of r
	A1 cso	For the correct required expression of $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ (must have $\boldsymbol{U_n}$)
(b)	B1	For stating the correct reason, $\lvert r\rvert < 1$ or -1 < r < 1 or "0.8" < 1 or $\left\lvert "\frac{4}{5}" \right\rvert < 1$ **Do not accept** just r < 1 without a correct reason
(c)	M1	For using their value of A /a /first term of their geometric sequence and their r (provided their $\lvert r\rvert < 1$)in a correct formula for the sum to infinity. $S_\infty = \frac{'their\ A'}{1-'(their\ r)'}$
	A1cao	For the correct value of 1000
(d)	M1	Uses their a and r (their r can be $\lvert r\rvert < 1$ or $\lvert r\rvert > 1$) to set the correct formula for the sum of a geometric series > 978 or =978 or ≥ 978 (>978 or =978 or ≥ 978 may be implied by later work.)
	dM1	Attempts to rearrange and achieves $(their\ r)^n$ " < "k (k *is non zero positive constant*) **Allow strict or non-strict inequalities or equation**: <,>, ≥ , ≤ or =
	ddM1	For correctly takes logs (any base)/ln, and achieves a positive value for n, it maybe implied by 17.1…but not 18 **ALT**: trial and error to find a value of n (at least tries n=17 and n=18)
	A1	For n = 18 Must come from **correct** working, achieves n=18 with no incorrect inequalities in their working.

If a candidate does not score the dM1 mark but arrives at an answer of 17.1... this scores a maximum of M1 as clear use of calculator is not condoned.

Type MS · Does a geometric series have a sum to infinity?

Answer 3 of 4 · Question 8 part b · 1 mark

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Does a geometric series have a sum to infinity?

Answer 4 of 4 · Question 6 part b · 1 mark

2025 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

6(i)(a)	$\left[\frac{ar}{a} = r = \right] \frac{\sqrt{3}}{3+\sqrt{3}} \times \frac{3-\sqrt{3}}{3-\sqrt{3}}$ oe or $\frac{\sqrt{3}\left(3-\sqrt{3}\right)}{3^2-\left(\sqrt{3}\right)^2}$ $= \frac{3\sqrt{3}-3}{6}$ oe $\Rightarrow \frac{\sqrt{3}-1}{2}$ or $\frac{-1+\sqrt{3}}{2}$	M1 A1 [2]
(b) (i)	$\frac{\sqrt{3}-1}{2}\left(=0.366\right)<1$ or $\lvert r\rvert<1$ so the series is convergent	 B1ft [1]
(ii)	$S = \frac{8\left(1-0.6^n\right)}{1-0.6}$, $T = \frac{8}{1-0.6} = [20]$ $\left["20"-"20"\left(1-0.6^n\right)<0.12\right] \Rightarrow 0.6^n < 0.006$ oe eg $-0.6^n > -\frac{3}{500}$ $n\log_{10} 0.6 < \log_{10} 0.006$ oe eg $n\log_{0.006} 0.6 < \log_{0.006} 0.006$ $n > \frac{\log_{10} 0.006}{\log_{10} 0.6}$ $\left(n > 10.015\right)$ $n = 11$	M1, M1 M1 dM1 A1 [5]
		Total 8 marks

Type MS · Does a geometric series have a sum to infinity?

Answer 4 of 4 · Question 6 part b · 1 mark

2025 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For dividing the second term by the first term and for a correct method **stated** to rationalise the denominator. A correct answer may not imply this mark as candidates are told to show working.
	A1	For $\frac{3\sqrt{3}-3}{6}$ oe **and** $\frac{\sqrt{3}-1}{2}$ or $\frac{-1+\sqrt{3}}{2}$ Accept $p = -1$ and $q = 2$ Minimum steps as shown in the mark scheme needed, with no errors or omissions. Additional steps shown must be checked and correct.
If students start completing 1st term divide 2nd ie $\frac{3+\sqrt{3}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{3}+3}{3}$ they must progress to $\frac{3}{3\sqrt{3}+3}$ oe eg $\frac{1}{\sqrt{3}+1}$ before marks can be awarded.		
(b) (i)	B1ft	For a correct explanation referring to r or the value being < 1 or $\lvert r \rvert < 1$ This may ft any value of r from their part (a) where $-1 < r < 1$ Accept "r is between 1 and – 1"
(ii)	M1	For the correct expression for S or T using the given values. **May be implied/embedded within the solution.**
	M1	For the correct expression for S and T using the given values. **May be implied/embedded within the solution.**
	M1	For an inequality of the form $0.6^n < A \quad A > 0$ or $-0.6^n > -B \quad B > 0$ $0.6^n < 0.006$ oe eg $-0.6^n > -\frac{3}{500}$ will definitely imply first 2 M marks if one or both are not explicitly stated.
	dM1	For using logarithms correctly, including reversing the inequality to find a value for n Dependent on previous method mark
	A1	For the correct value of n **following correct work.** Candidates must use logarithms and we must see all previous method marks. If ($n = 11$) comes from incorrect logs work or incorrect reversal of this inequality, this will be A0.

Type MS · Does a geometric series have a sum to infinity?

Answer 4 of 4 · Question 6 part b · 1 mark

2025 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks